

Technology Stack

Date	05 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js (Vite), React Router, Axios, Recharts	Fast, component-based UI with charting for financial dashboards.
2	Backend / Server-Side	Python + FastAPI, SQLAlchemy, Pydantic	High-performance async REST API with strong data validation and ORM.
3	Database / Data Storage	SQLite (easily configurable for PostgreSQL)	Lightweight for development; scalable to PostgreSQL for production.
4	Cloud / Hosting / Deployment	Render (Frontend + Backend)	Simple CI/CD-friendly hosting for both services.
5	Version Control & CI/CD	Git & GitHub	Version control and project history tracking.
6	Third-Party APIs / Other Tools	Google Gemini API, ReportLab, JWT (python-jose), bcrypt	AI-powered financial insights/chat, PDF report generation, and secure authentication.